Software Engineering

CSC648/848

Henry
VILLAR

SW PROJECT MANAGEMENT

SW PM, Ethics and IP

Important

SW Project Management – Key goal

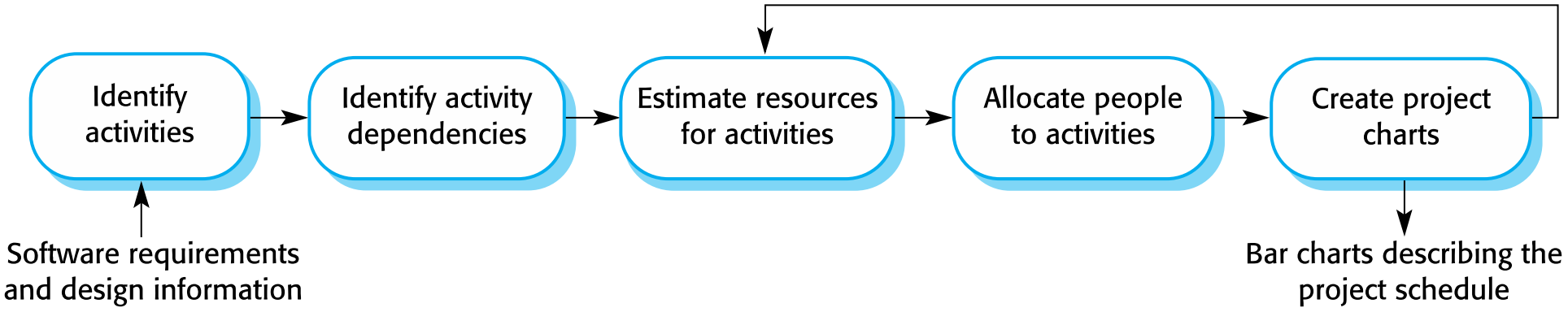
Goal: Build and deliver the SW product that:

1. Satisfies **specs** (no more, no less)
2. It's done on **budget**
3. It's done on **schedule**
4. Delights the **customers**
5. Is **maintainable**, easy to sell, well supported, documented and marketed…

" Good is better than the best"

Important

The project scheduling process



Progress Meeting (Video): https://youtu.be/u8Kt7fRa2Wc

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Important

Task duration and definition

- Initially, allocate at most weekly granularity for tasks, later use short duration (usually down to 2 days)
- If task is more than 8 weeks, set a checkpoint/milestone in between
- Initially have weekly check procedure
  - Progress report to manager, rolled up
  - Meeting
- More often would add too more overhead
- Exceptions in time of crises, resolving risks, close to delivery
- Even daily meetings can be effective if done properly
- Report on progress by all (go around the table):
  - On track or not, issues, risks

Important

Duration and resource estimation

- Usually provided by SW developers or tech leads
- Various people have different ability to estimate: some always underestimate. Know this and readjust for each person
- Use previous experience (good or bad)
- Do not assume everything will be fine
- Principal resource in SW development is human time
- The more novel, the more risky ➔ readjust the schedule
- Do not forget unit testing, integration testing and documentation
- Rule of thumb: whatever people tell you, multiply with 1.3 or 1.5 safety factor

Important

Mythical man-month syndrome

- Coined by Brooks (at times where mostly men were doing SW)
  - Adding people late in the project is counterproductive: people training and integration effort is most often more than the original delay
- Think how long it takes to get a new person to be productive – easily 2 months…
  - Develop the job adv. and get management approvals
  - Advertise
  - Collect resumes and evaluate
  - Interview
  - Bring on board
  - Train

Important

Ways to present project schedules

- Bar charts
- Activity networks
- Gantt chart (e.g. implemented in Microsoft Project)

Can be done in simple document editors or special tools (Simple, no special tools is OK and desired)

Important

Task durations and dependencies

Task	Effort (person-days)	Duration (days)	Dependencies
T1	15	10	
T2	8	15	
T3	20	15	T1 (M1)
T4	5	10	
T5	5	10	T2, T4 (M3)
T6	10	5	T1, T2 (M4)
T7	25	20	T1 (M1)
T8	75	25	T4 (M2)
T9	10	15	T3, T6 (M5)
T10	20	15	T7, T8 (M6)
T11	10	10	T9 (M7)
T12	20	10	T10, T11 (M8)

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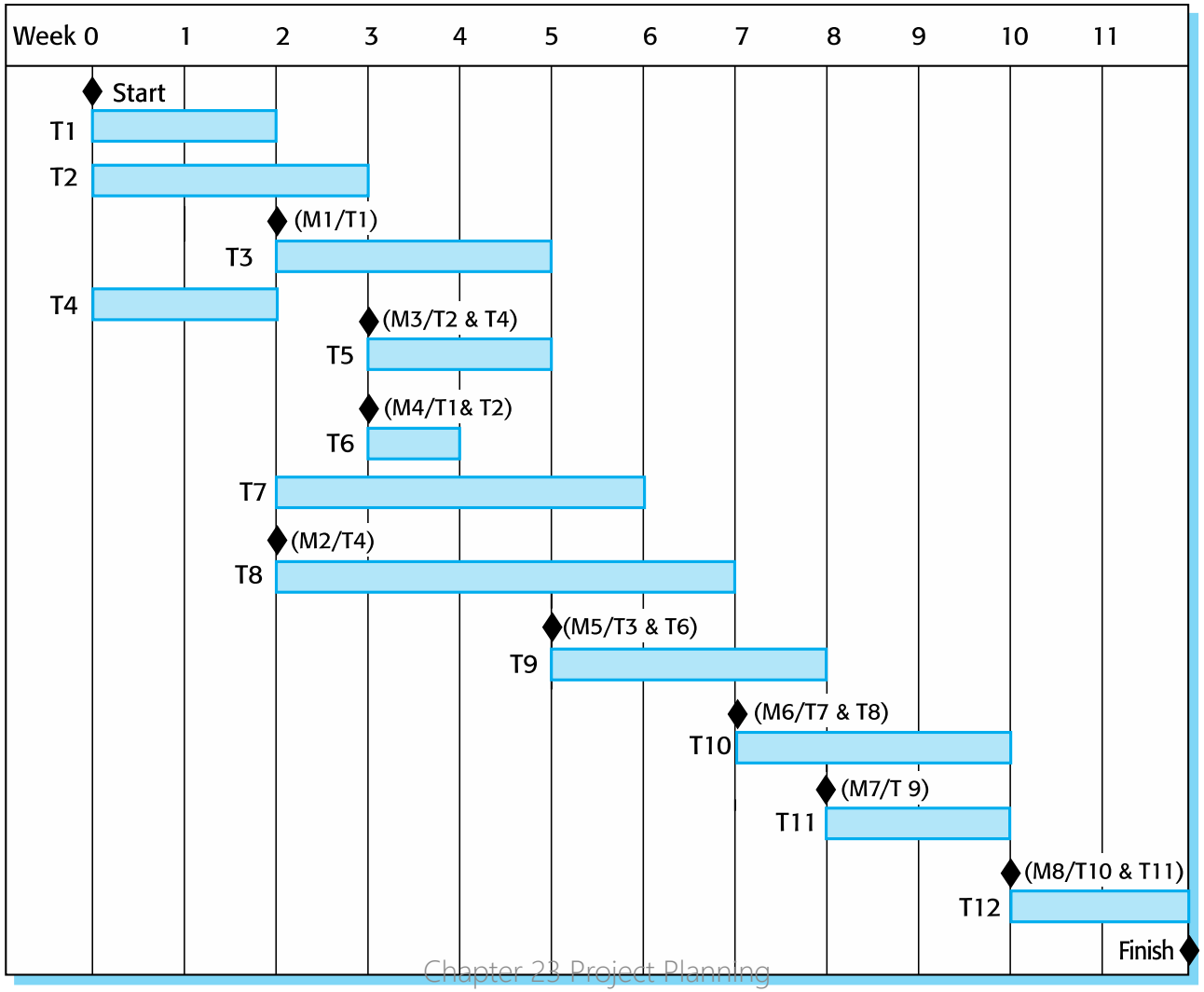
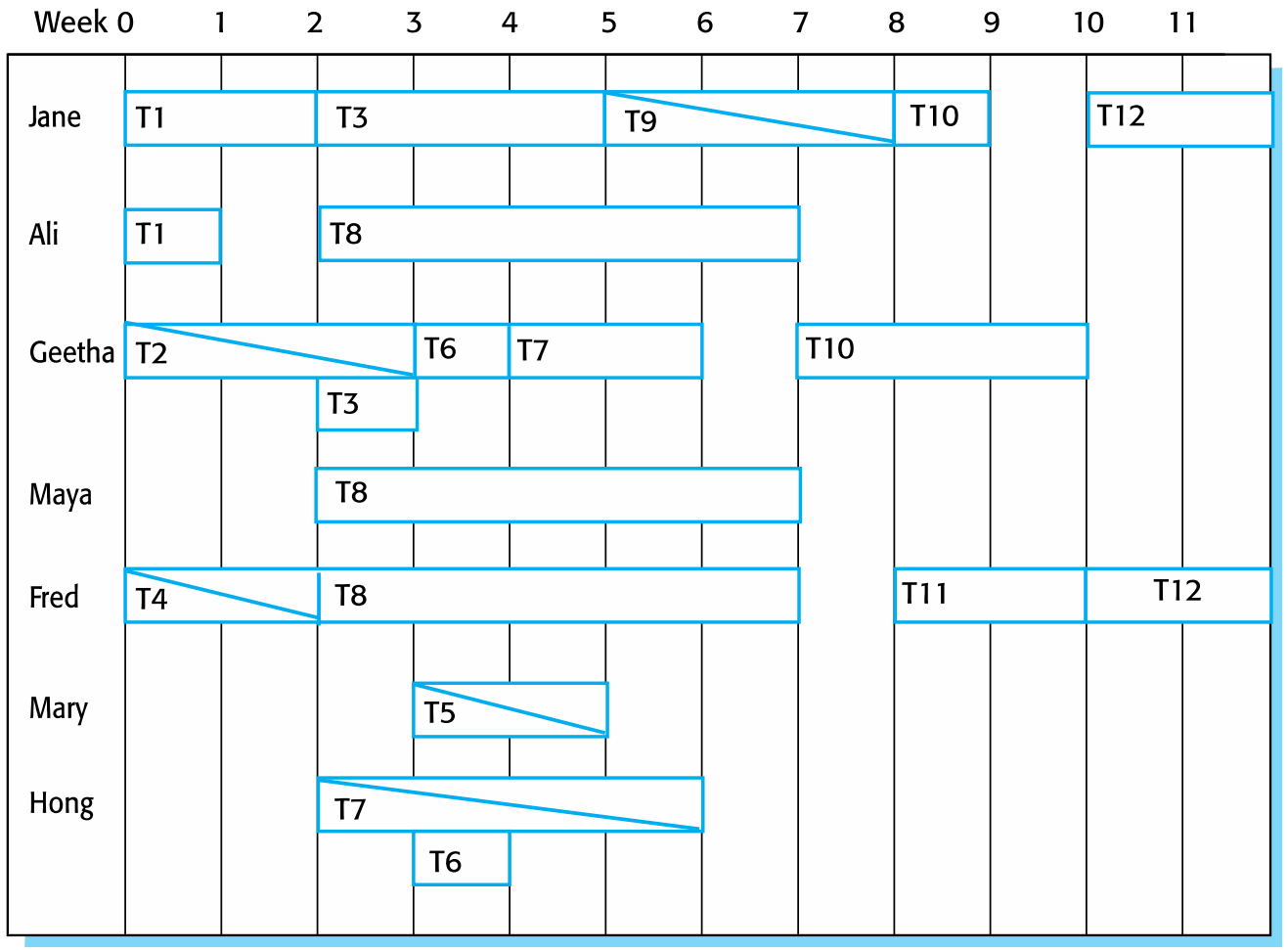
Important

Activity bar chart (Gantt chart)



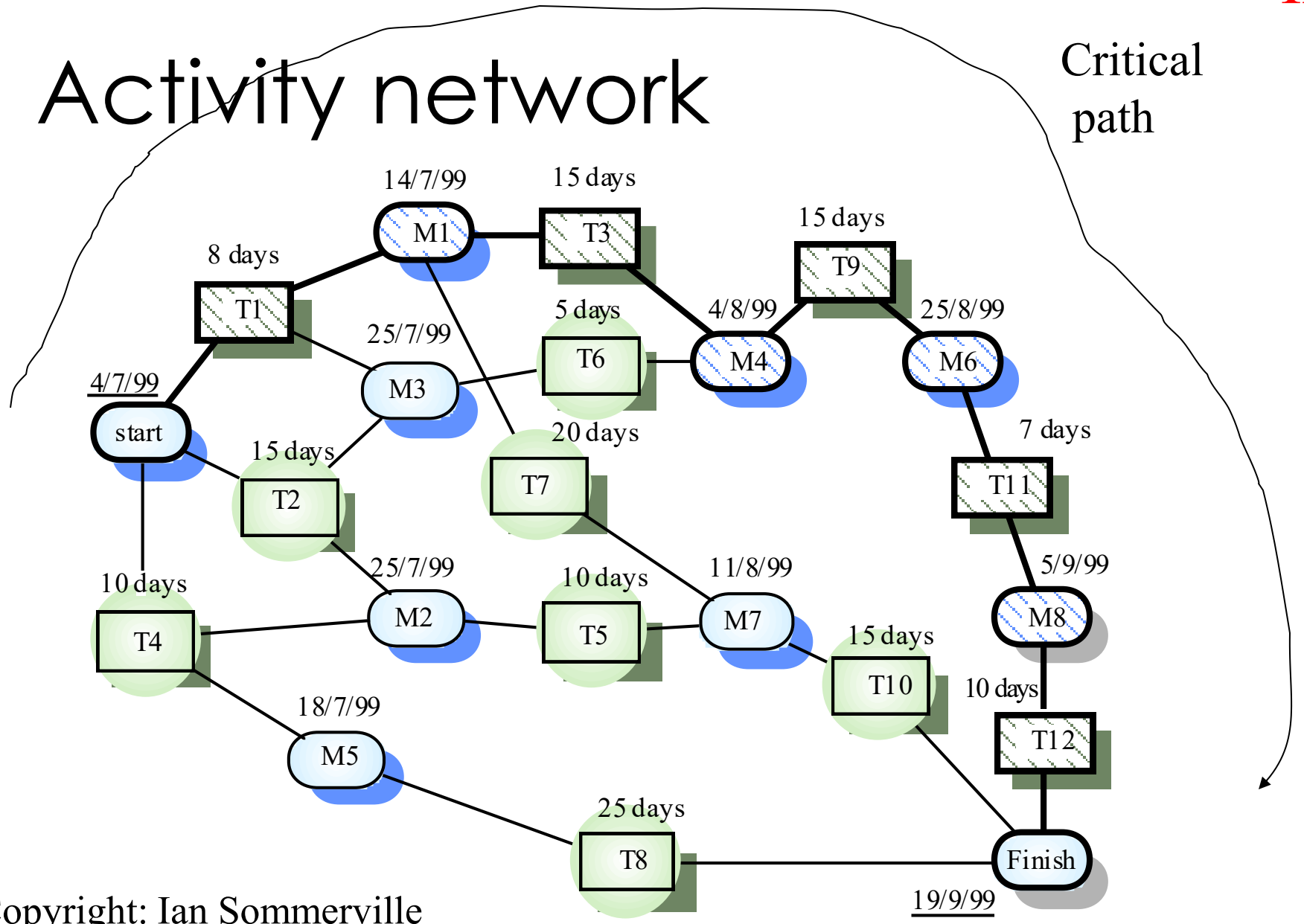
Important

Staff allocation chart



Important

Activity network



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Important

Dependencies

- Often, for one task to start, one or more tasks have to be completed first ➔ dependencies
- Milestones have multiple task dependencies ➔ all those tasks have to be completed before milestone can be reached
- Critical path: the longest path in the activity charts ➔ it determines the duration of the projects
- Delays in critical path task cause delay in the project
- Delays in non-critical path tasks may be tolerated as log as they are less than critical path

Important

Issues with SW Requirements

- Seldom in writing, often verbal
- Vague, many implicit ones
- May be conflicting
- Change a lot (for good and bad reasons)
- Hard to know/understand/comprehend all of them up front
- …

Important

Planning, cost estimation, PM

- Usually, 90% of the cost is people
- Programmer productivity can vary 10 X
- Great deal of people management needed
- Adding people does not help (even can create problems - Mythical man-month" by Brooks)
- Very difficult to monitor progress

- BUT: SCOPE can be changed even at the end (e.g. drop a function)

Important

Tasks of SW Manager

- Development of product vision and goals, and communicating them
- Develop a Plan – schedules, budget, resource needs and what is delivered (for the team and for the management)
- Execute, manage to successful delivery
  - Proposal writing and presentations
  - Funding and budget management
  - Planning
  - Scheduling
  - Costing
  - Providing adequate resources
  - Dealing with all stakeholders inside and outside the company
  - Monitoring and tracking
  - Reviews
  - Help resolving technical conflicts
  - Resolving personality conflicts
  - Develop vs. buy vs. vend out decisions
  - Report writing and presentations
  - Dealing with global development

Multiple communication Needed for SW management

Important

Important

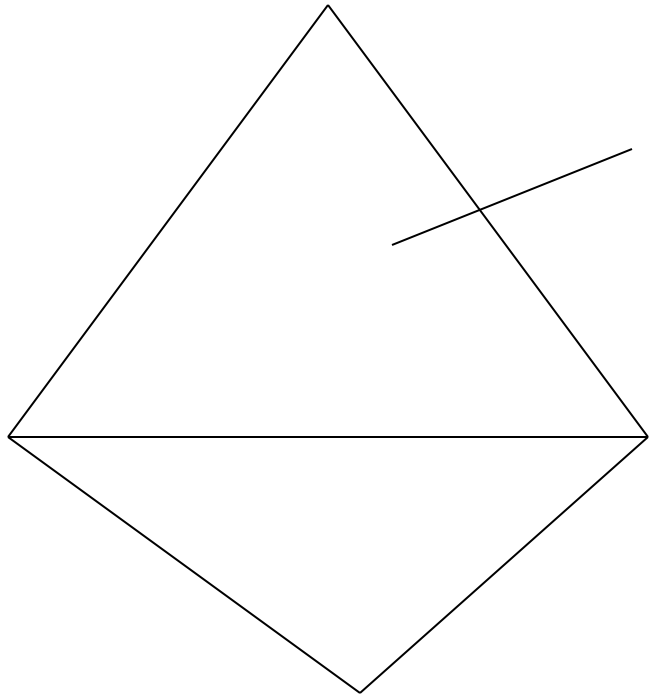
Project planning

- Project planning is iterative, ongoing activity:
- Start form best plan you can get
- Iterate, change, adjust often (more often in times of crisis or delivery) based on ALL pertinent information
  - Technical
  - Internal (progress of the team etc.)
  - External (changes in user requirements, competition, market)
- Keep overall goals in mind (delivery, schedules, budget and users)
- Drastic changes in plan need to be approved and communicated (I.e. schedules are there to me met)
- Remember: requirements change, and it is hard to monitor progress, so be nimble and iterative and check/revise often ➔ adopt agile SE processes

Variables in planning - they are related

Important

Cost (people, resources)



3 TRADITIONAL LEVERS TO PLAY WITH

Quality

Time (schedule)

SCOPE (features)

(E.g. if you want more features
It either requires more people
OR more time OR quality needs to
be reduced)

Unique to SW

Important

Vertical vs. matrix management

- *Vertical*: all resources are in one department.
  - Pros: easier to organize
  - Cons: duplication, not optimal use of resources (I.e. repeated UI development team for each product line)
- *Matrix*: Resources marshaled on per project basis from different departments
  - Pros: better use of resources, optimal talent used
  - Cons: harder to manage and organize. Critical to have cooperation between project manager in charge of project and other managers contributing resources

Important

Checkpoints, Milestones and deliverables

- SW is hard to "touch and see", it is intangible ➔ some verifiable and well defined milestones are needed for monitoring the progress

- *Checkpoint*: a defined time you will check some specific smaller issue (used internally to resolve technical or other problems)
- *Milestone*: end point of some logically distinct and defined activity: report, mockups, algorithmic solution, DB connected to UI etc.
- *Deliverable*: milestone that is delivered to somebody outside of the project, can be verified and measured. Often has contractual importance.

- Deliverable is also a milestone, but milestone may not be a deliverable
- M1-4 are milestones, M5 is deliverable

- Spread milestones/checkpoints in time, at least every 2-3 weeks)

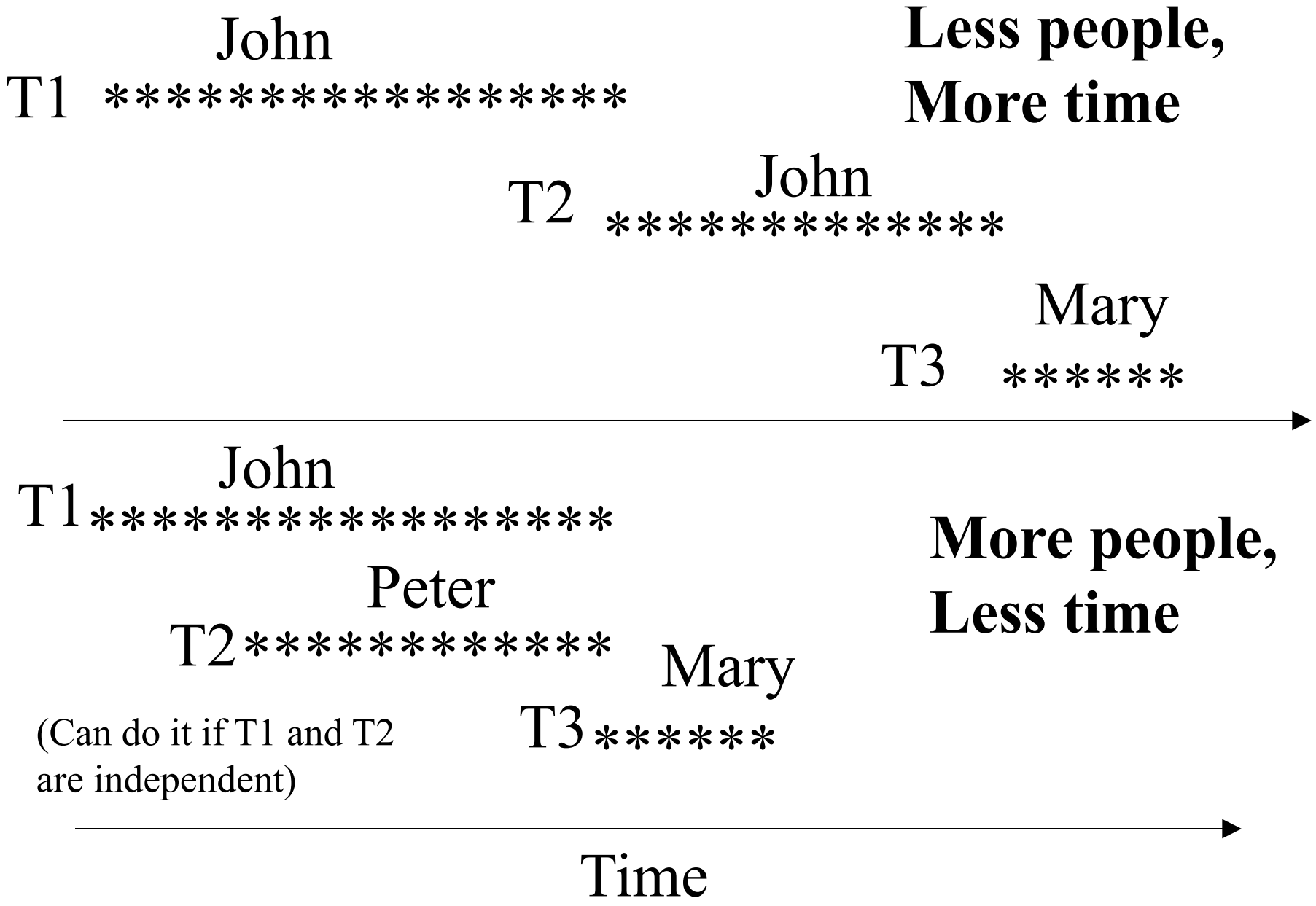
Simple way to do Gantt chart – no tools

```
        Jan    Feb    March  April

Task 1  ************************
Task 2                 **********
Task 3         *********
```

Revisions/Optimization

- Task scheduling and people allocation can be modified in some cases to reduce critical path or costs
- Perform activities in parallel (need people)
- Serialize non critical activities as long as critical path is the same (save people)


John
T1 **********
Less people,
More time
John
T2 **********
Mary
T3 ******
John
T1**********
More people,
Less time
Peter
T2**********
Mary
T3******
(Can do it if T1 and T2
are independent)
Time

Important

1) Use project requirements and design documents (tasks, UI, use cases, architecture) + budget, resources and completion date to get basic info and project parameters and constraints (I.e. Final Project M1 and M2)

2) Define project tasks, milestones and deliverables (variables: cost, time, quality, scope)

Only if critical problems arise

3) Define dependencies

4) Estimate task duration

5) Estimate task resources (people, equipment, SW, HW)

6) Draw up project schedule (I.e. backproject from desired completion date. IF big changes needed go back to Step 1)

7) Review periodically (more often toward the end of the project or when risks are identified)

8) Revise as necessary

9) IF problems arise initiate review

10) Loop back to Step 6 until project done

Summary: Steps in project management

SE ethics

- You are the key developer of a SW for a medical system, the SW has to be safe and reliable, patient lives are at risk
- Your boss asks for tight schedule, pushes aggressively for project to be completed
- You worry that the resulting SW is not safe for patients
- You are afraid you might be fired if you resist your boss

- What do you do? ➔ Apply SE ethics

SE ethics

- What to do?
  - Refuse to compromise
  - Compromise
  - “Blow the whistle”

- Difficult, and legally not fully regulated for now

Ethical AI – BIG topic of today

- AI is becoming part of many critical applications in health, military business and everyday life (driverless cars, loan applications)
- BUT:
  - Do we know HOW it works (Explainability)
  - Is it tested well, is it reproducible and verifiable
  - Is it designed ethically to be safe and non-discriminatory
  - If it causes problems who to complain to, can you sue for damages (and who to sue)
- What do you do as developer and tester of AI systems

Readings on AI Ethics – really important stuff to read and know – big topic

- AI Principles – Future of Life Institute
  - https://futureoflife.org/ai-principles/

- Upcoming Workshop on AI Ethics in Biomedicine, D. Petkovic et al, PSB 2020
  - https://psb.stanford.edu/workshop/ai/
- EU General Data Protection Initiative (GDPR)
  - https://en.wikipedia.org/wiki/General_Data_Protection_Regulation

SE Ethics and IEEE/ACM Society

- IEEE Computer Society and ACM approved SE code of ethics 1999:
  - https://www.acm.org/about-acm/code-of-ethics
- Still legally not binding

IEEE/ACM code of ethics V 5.2 - summary

Important

1. PUBLIC - Software engineers shall act consistently with the public interest.
2. CLIENT AND EMPLOYER - Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. PRODUCT - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. JUDGMENT - Software engineers shall maintain integrity and independence in their professional judgment.
5. MANAGEMENT - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. PROFESSION - Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. COLLEAGUES - Software engineers shall be fair to and supportive of their colleagues.
8. SELF - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Important

Some other things you need to know – Intellectual Property (IP)

- Copyright
- Trademarks
- Patents
- Trade secrets
- SW licensing

Note: critically important in USA – has legal and criminal consequences

Important

Copyright (SW)

- Copyright protects the expression of an idea, not an idea itself – defines the exclusive owner
- Used for images, text, music, SW code
- Gives rights to the owner and protects from unauthorized copying – use requires permission of the owner
- (Does not define the distribution – that is done by licensing)
- See http://en.wikipedia.org/wiki/Copyright
- SFSU students own copyright to the SW and any material they produce as part of their student work at SFSU

Copyright of digital media

- Digital media (images, video, sound..) is usually copyrighted and you can not copy it and redistribute
- In some cases “fair use” is allowed
- Need permission and agreement with others to post, distribute.
- http://en.wikipedia.org/wiki/Digital_library#Copyright_and_licensing
- http://en.wikipedia.org/wiki/Licensing

Important

Trademarks

- Used to protect distinct logo, sign, word, phrase specific to business or service
- Can be registered or unregistered
- See http://en.wikipedia.org/wiki/Trademarks

Important

Patents

- Protect an idea, not particular implementation or embodiments
- Covers for e.g. algorithm, methods, processes, medical drugs
- Granted for period of time to inventors/organizations
- Must be applied for
- Check http://en.wikipedia.org/wiki/Patent
- Familiarize yourself with how patent looks https://patents.google.com/patent/US6182069

Important

Trade secrets

- Method, formula, process kept secret by the company (and not patented) to offer it competitive advantage (note: patents expire after some time).
- Coca Cola
- Check http://en.wikipedia.org/wiki/Trade_secrets

Important

SW licensing

- SW license is legal way of covering its distribution (different from copyright which covers the ownership/authorship)
- Commercial e.g. when purchasing SW (EULA's – end user licensing agreement)
- Open Source allow free distribution (BUT check for details – is the SW which uses Open Source also required to be free?)
- Check http://en.wikipedia.org/wiki/Software_license

Important

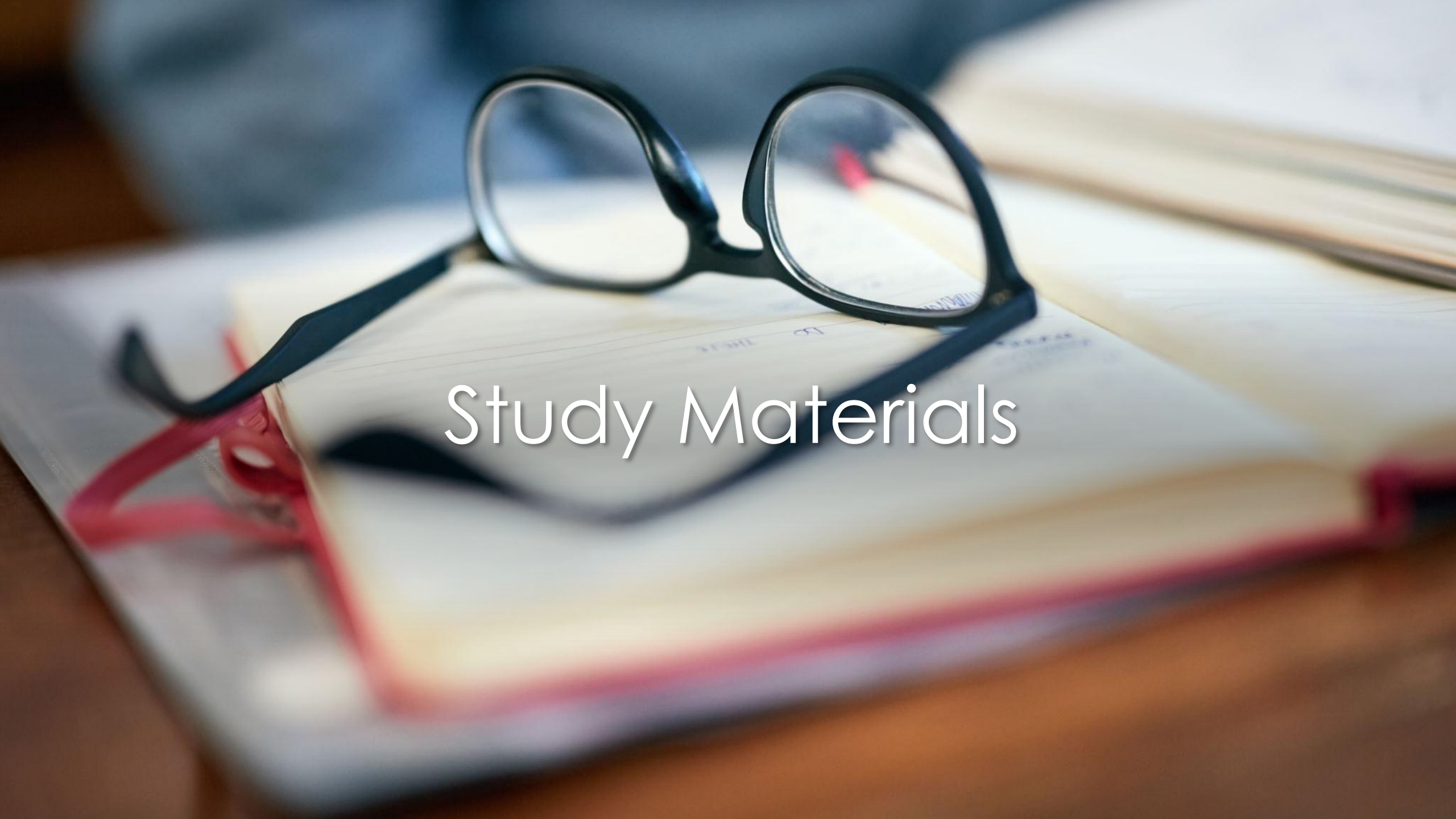
IT usage and SW licensing at SFSU

- Familiarize yourself:
  - http://cs.sfsu.edu/student-software-licensing-policies
  - http://cs.sfsu.edu/it-resources-acceptable-use-policy

Team Project Time

- M2 **More Detailed Requirements, Specs, Architecture, UI mock-ups and Vertical SW prototype -** Two goals:
  - ***More detailed design - Milestone 2 document***: prioritize requirements; design high level UI mockups and storyboards; design high level DB organization, and practice UML. *(one per team, submitted similarly like Milestone 1)*. Use as much space as you need, but the expected length is about 20-25 pages. You must submit <u>all</u> sections.
    1. Functional Requirements – prioritized
    2. <u>UI Mockups and Storyboards (high level only)</u>
    3. High level Architecture, Database Organization
    4. High Level UML Diagrams
    5. <u>Identify *actual* key risks for your project at this time</u>
    6. <u>Project management</u>

  - ***Vertical SW Prototype***: Develop first simple bare-bones prototype (from DB to browser) to test the infrastructure, educate the team, resolve technical issues and also serve as basis of further development (must run on the deployment server). *(one per team) to be reviewed in class and also by Class CTO (off-line)*



Study Materials

SW Management - Wikipedia

- http://en.wikipedia.org/wiki/Software_project_management

SW management is also (from Wikipedia link above):

- Risk management is the process of measuring or assessing risk and then developing strategies to manage the risk. plan.
- Requirements management is the process of identifying, eliciting, documenting, analyzing, tracing, prioritizing and agreeing on requirements and then controlling change and communicating to relevant stakeholders.
- Change management is the process of identifying, documenting, analyzing, prioritizing and agreeing on changes to scope (project management) and then controlling changes and communicating to relevant stakeholders.
- Software configuration management is the process of identifying, and documenting the scope itself, which is the software product underway, including all sub-products and changes and enabling communication of these to relevant stakeholders.
- Release management is the process of identifying, documenting, prioritizing and agreeing on releases of software and then controlling the release schedule and communicating to relevant stakeholders

Milestones and deliverables: examples

- 80% complete is <u>not a good milestone</u> – hard to verify and measure for SW projects
- Good ones:
  - Architecture agreed upon
  - Data access designed, implemented and unit tested
  - UI mockups done
  - Check that Google maps can be integrated in our system – throw away prototype
  - DB connected with front end, works according to specs
  - Interface defined, unit tested and implemented
  - User study done
  - Images scanned and imported
  - Beta delivered to customers for evaluation

Scheduling – it is hard…

- SW in intangible with progress hard to monitor (vs. building a house)
- Technical uncertainties
- Great variation in programmer productivity (over 10 X) (vs. say windows installers)
- Hard to reuse previous examples (but OK as a starting point)
  - Tools and technology change
  - Different people have different productivity
- Risk with use of latest technology (teams needs to learn, bugs, incomplete features)
- Unanticipated changes: customer requests, requirements, business, market, competition

Very difficult and demanding ➔ requires experience and <u>constant iterations and reevaluation</u>

SW Project Management - resources

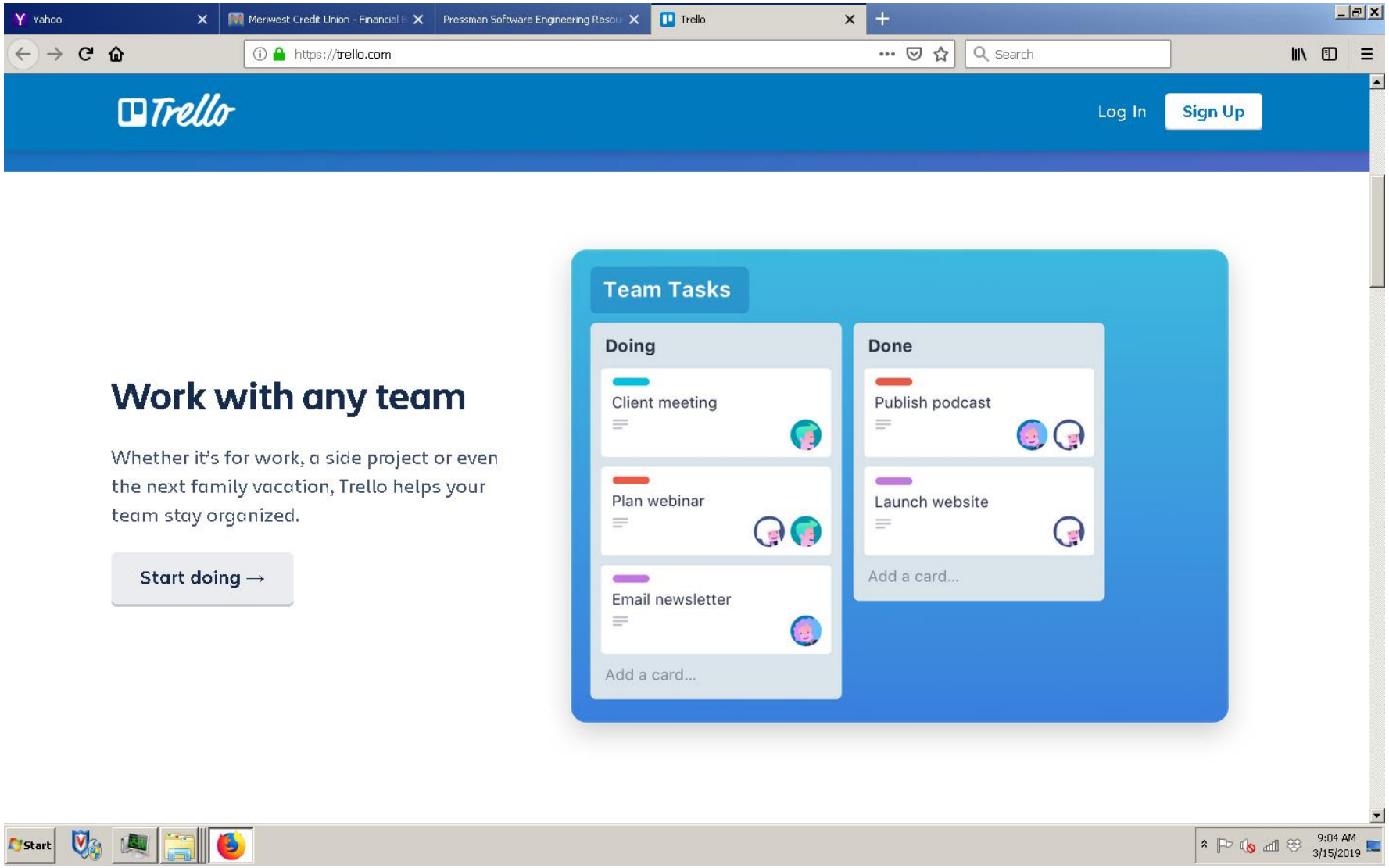
- List of some good resources (many available)
  - http://www.rspa.com/spi/project-mgmt.html#resources
- Lately, a lot of tools emerged to help manage tasks list in a teams setting, e.g.
  - Trello
    - https://trello.com/

Important for class team leads

- For every task you assign have a person responsible and the deadline
- For every person have a task

- Check progress around middle of the time line
- Deadlines are to be respected!
- Use task management systems that offer shared dashboard (e.g. Trello)

Trello


Yahoo
Meriwest Credit Union - Financial
Pressman Software Engineering Resou
Trello
https://trello.com
Search
Trello
Log In
Sign Up
Work with any team
Whether it's for work, a side project or even the next family vacation, Trello helps your team stay organized.
Start doing →
Team Tasks
Doing
Client meeting
Plan webinar
Email newsletter
Add a card...
Done
Publish podcast
Launch website
Add a card...
Start
9:04 AM
3/15/2019

Case study

- You are PM for a startup which developed a new product
- You developed a new logo for the company
- You developed novel alg. for search/data mining that is your key advantage
- You will develop APIs to distribute and sell
- You need to use some digital images made by others for your WWW site

- Question: What IP protections you need for each?

Review Questions

- Why is SW hard to manage and why is it unique
- What are some of the typical tasks for SW manager
- What is the importance of people management in SW, and why
- If your SW is built by many teams globally located, what are some issues you have to pay particular attention in terms of final delivery, management of the whole process etc.
- What are variable a typical SW manager can work with in scheduling and planning

Review Questions

- Vertical vs. matrix management
- What are milestones and what are deliverables
- What is critical path
- Draw typical bar charts, activity diagrams, Gantt charts
- What are typical organizations of SW teams
- Go over cases given in the class slides and think what would you do as manager, manager's manager or SW developer
- Know SE ethics principles as approved by IEEE/ACM
- Know various IP issues and what works in what acse

Summary

- Issues that need utmost care:
  - High level project management, organization and coordination
  - Centralized decisions and coordination for key high level issues (i.e. requirements,specs, architecture)
  - Communication: teamwork, trust, written and verbal communication, conflict resolution
  - Culture
  - People management
  - Tools: for communication, project management, SW development
  - Technical management:
    - Task allocation vs. modular architecture, well defined interfaces and functions etc.
    - Source control and management
    - QA and integration testing
    - Lower-level project management including risk management
    - Encouraging best ideas and innovation while promoting teamwork

Thank You